SAR OSC Test: Python (Mac) → TouchDesigner (Windows) over Ethernet

v0.1 experimental · 2026-07-29 · not yet run

Goal: get a Mac running Python talking to a Windows box running TouchDesigner over a dedicated Ethernet link, using OSC. This is just a wiring/comms test — it's deliberately outside the SAR pipeline (no L0-L5 stuff here). Once it works reliably, it becomes the foundation for sensors, middleware, agents, etc.

Steps: wire it up → confirm ping → confirm OSC receives → then (later, separately) hook it into TouchDesigner for real.

Not covered here: SAR ingest normalization, pipeline conformance, semantic processing — that's a separate phase, after this passes.

Hardware

Device	Role
Mac	Python dev
Windows PC	TouchDesigner
NETGEAR GS316EPP	Gigabit switch
2× Cat6	Dedicated link

Both machines stay on Wi-Fi for internet; Ethernet is just for SAR control traffic.

```
Mac (192.168.50.10) — GS316EPP — Windows (192.168.50.20)
```

(Wi-Fi handles internet on both machines separately, in parallel.)

Why Ethernet: lower latency, no dropped packets from Wi-Fi congestion, predictable timing, easier to debug. Keeping it a separate physical network keeps control traffic off the internet-facing connection entirely.

Network settings

Network	192.168.50.0/24
Mac	192.168.50.10
Windows	192.168.50.20
Gateway / DNS	leave blank on both

No gateway set = this interface is never used for internet traffic, so Wi-Fi keeps handling that automatically.

Mac setup

1. Plug in Ethernet.
2. System Settings → Network → select the Ethernet adapter.
3. Set IPv4 to Manual: `192.168.50.10`, mask `255.255.255.0`, gateway/DNS blank.
4. Apply, then check with `ifconfig` that it's active with the right IP.
5. Confirm you can still get online over Wi-Fi.

Windows setup

1. Plug in Ethernet.
2. Settings → Network & Internet → Advanced network settings → More adapter options → right-click Ethernet → Properties → IPv4.
3. Set static IP `192.168.50.20`, mask `255.255.255.0`, gateway/DNS blank.
4. Apply, check with `ipconfig` that it's up with the right IP.
5. Confirm Wi-Fi internet still works.

Check the link

bash

```bash
# from Mac
ping 192.168.50.20
```

cmd

```cmd
:: from Windows
ping 192.168.50.10
```

Both directions should get clean replies, no dropped packets. If not — check the cable, switch port lights, adapter selection, and IPs before going further.

Install python-osc

```bash
python -m pip install python-osc
```

Set up TouchDesigner to listen

1. Windows Firewall: allow TouchDesigner on the private network profile (add an inbound UDP 9000 rule if needed).
2. In TouchDesigner: add an OSC In DAT, protocol UDP, local port `9000`, set it Active. Keep the DAT viewer open so you can see messages land.

First test

```python
from pythonosc.udp_client import SimpleUDPClient

client = SimpleUDPClient("192.168.50.20", 9000)
client.send_message("/hello", 1)
```

TouchDesigner should show `/hello 1`.

Stress test (recommended)

```python
from pythonosc.udp_client import SimpleUDPClient

client = SimpleUDPClient("192.168.50.20", 9000)
for i in range(500):
    client.send_message("/test/index", i)
```

Check that TouchDesigner receives all 500 in order — note anything missing or out of sequence.

What's next (after this passes)

Bidirectional messaging, heartbeats, a real OSC namespace, sensor integration, middleware, distributed services.

Done when

1. Ping works both directions, reliably.
2. TouchDesigner receives test OSC messages on the configured port.

3. The 500-message burst arrives with no unexpected loss.
4. Both machines keep internet over Wi-Fi the whole time.
5. Results are logged below.

Don't start SAR pipeline integration until this passes.

Run log (fill in during the actual test)

Environment: Mac OS ver. / Windows OS ver. / TouchDesigner build / Python ver. / python-osc ver.

Run: date/time — operator — Ethernet link status — ping results — OSC receive confirmed? — messages sent/received — loss/errors — internet-over-Wi-Fi retained (Y/N)

Outcome: Pass/Fail — notes

If something breaks

Check physical link before software, IP connectivity before OSC. Leave the Ethernet gateway blank so Wi-Fi stays the internet path. Use static IPs while you're still developing this.